Report: Geofence

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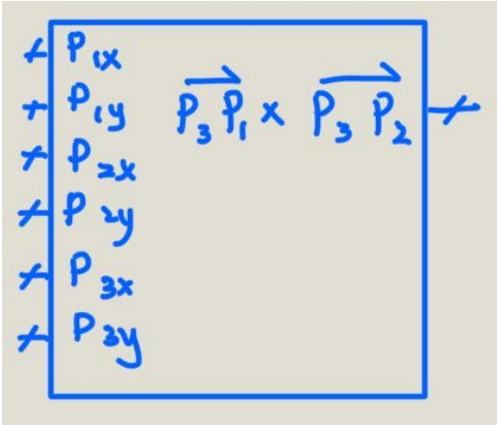
Date: 2026.3.29

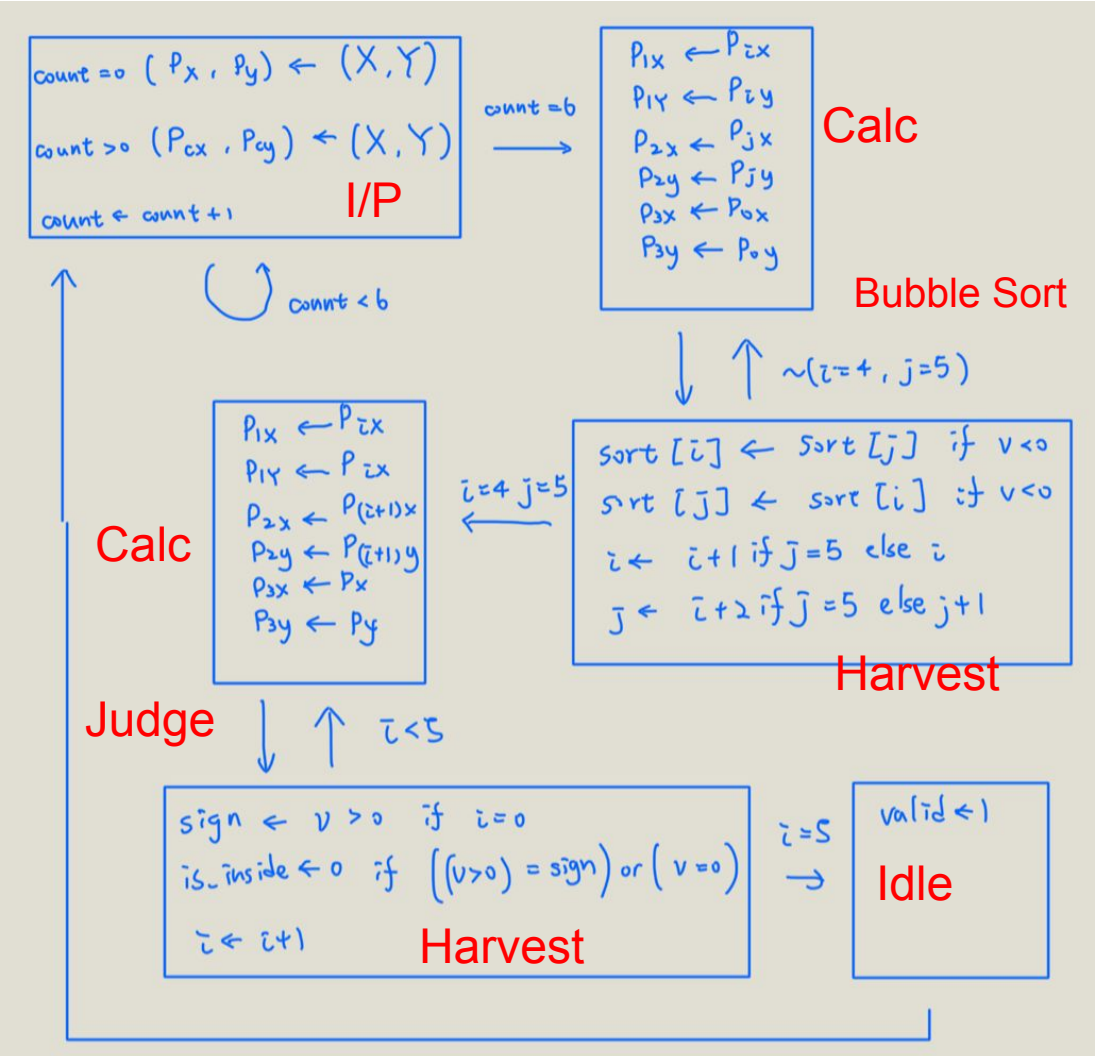
State Machine

Shared cross module

(A - B) x (A - C)

select input by MUX





Good-Result

Clock Cycle Time: 30 ns

Total Simulation Time: 60150 (ns)

Total cell area: 20149 (μm^2)

AT: 1.212 * 10^9 (ns-μm^2)

Level: A

area.log

```
Combinational area:              15594.014031
Buf/Inv area:                     1386.775776
Noncombinational area:            4555.821608
Macro/Black Box area:                0.000000
Net Interconnect area:          192920.970825

Total cell area:                  20149.835639
Total area:                     213070.806464
```

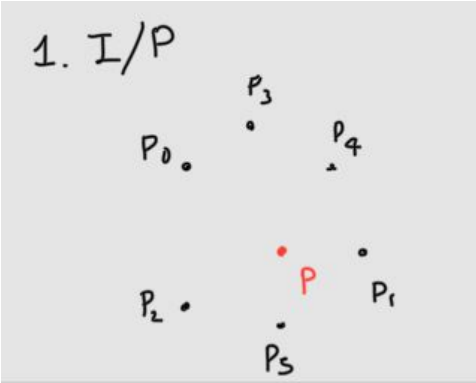
RTL Simulation Success

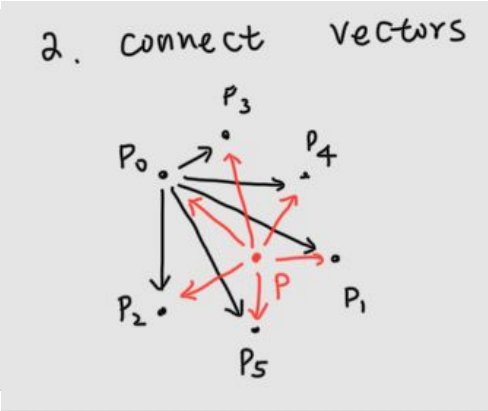
```
Scenario50(out):    X     Y
        Object:   321, 1000
           AP1:   321,  870
           AP2:   432,  123
           AP3:   970,  280
           AP4:   567, 1003
           AP5:   321,  127
           AP6:  1010,  643
Object50: Golde/Return => 0/0, PASS

----------------------------------------------
--    Simulation finish,  ALL PASS           --
----------------------------------------------
$finish called from file "tb.sv", line 149.
$finish at simulation time            6015000
           V C S   S i m u l a t i o n   R e p o r t
Time: 60150000 ps
CPU Time:      0.800 seconds;       Data structure size:   0.0Mb
Sun Mar 29 22:29:19 2026
CPU time: .788 seconds to compile + .657 seconds to elab + .711 seconds to link + .853 seconds in simulatio
```

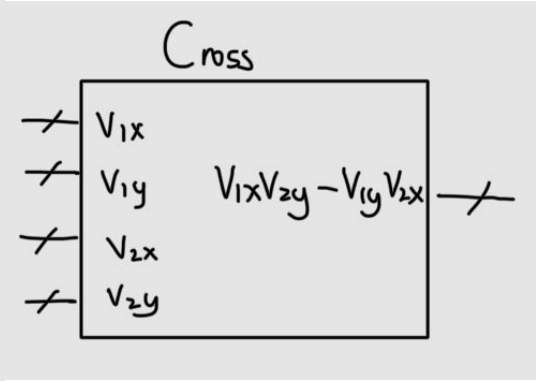
Gate Level Simulation Success

```
Scenario50(out):    X     Y
        Object:   321, 1000
           AP1:   321,  870
           AP2:   432,  123
           AP3:   970,  280
           AP4:   567, 1003
           AP5:   321,  127
           AP6:  1010,  643
Object50: Golde/Return => 0/0, PASS

----------------------------------------------
--    Simulation finish,  ALL PASS           --
----------------------------------------------
$finish called from file "tb.sv", line 149.
$finish at simulation time            60150000
           V C S   S i m u l a t i o n   R e p o r t
Time: 60150000 ps
CPU Time:      2.470 seconds;       Data structure size:   0.4Mb
Sun Mar 29 22:32:15 2026
CPU time: 1.818 seconds to compile + .798 seconds to elab + .741 seconds to link + 2.530 seconds in simulatio
```

Designed Method

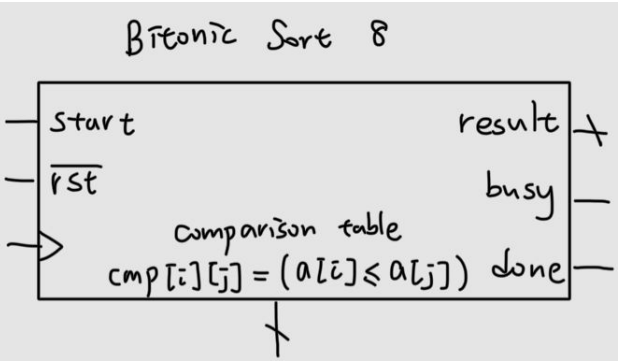


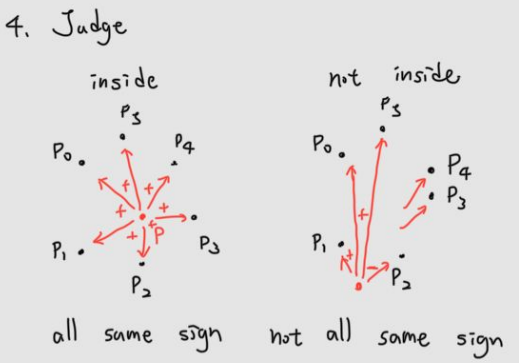




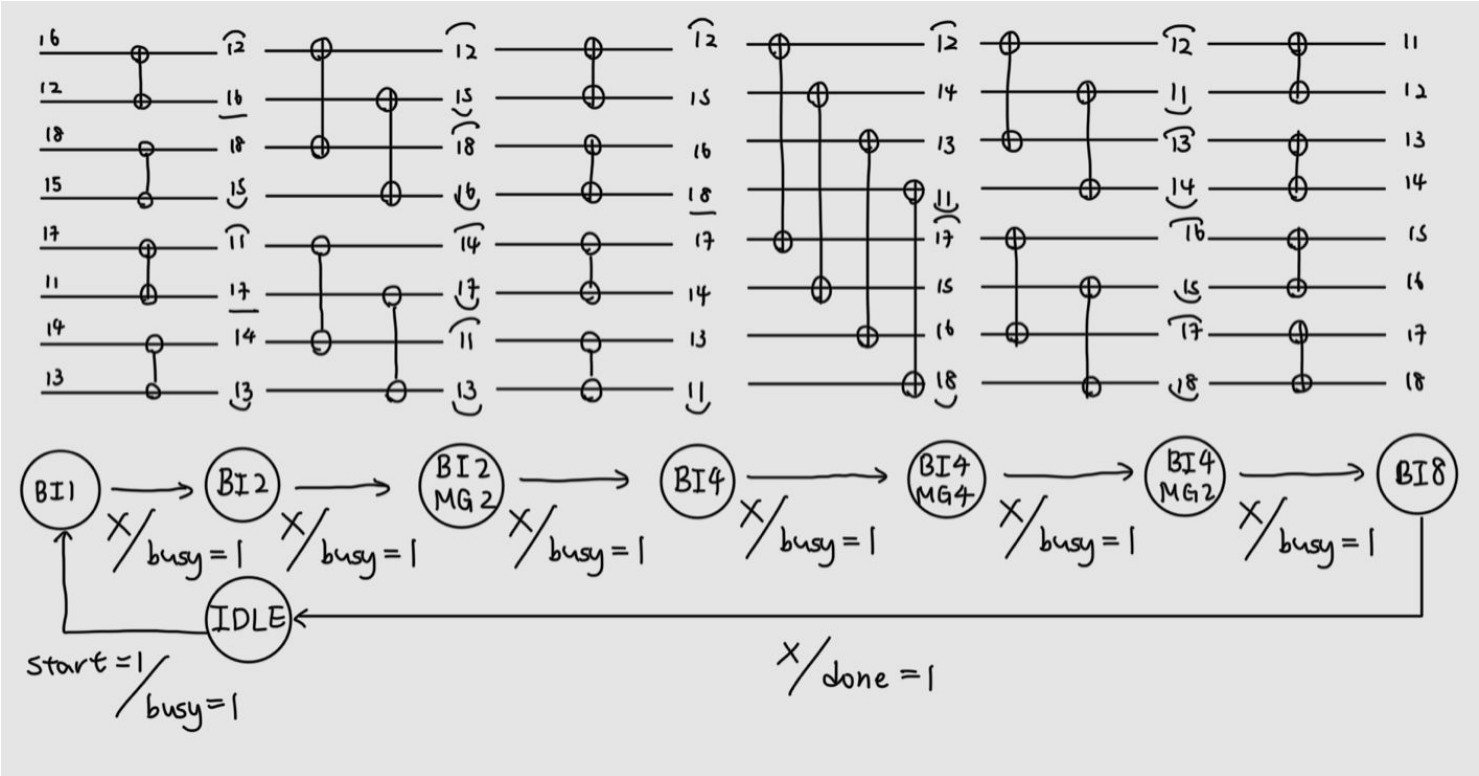
3. build comparison table

$\vec{a}\times\vec{b}$ ($\vec{a}$ \ $\vec{b}$)	$\vec{P_1P_0}$	$\vec{P_2P_0}$	$\vec{P_3P_0}$	$\vec{P_4P_0}$	$\vec{P_5P_0}$
$\vec{P_1P_0}$	X	<0	>0	>0	<0
$\vec{P_2P_0}$	>0	X	>0	>0	>0
$\vec{P_3P_0}$	<0	<0	X	<0	<0
$\vec{P_4P_0}$	<0	<0	>0	X	<0
$\vec{P_5P_0}$	>0	<0	>0	>0	X

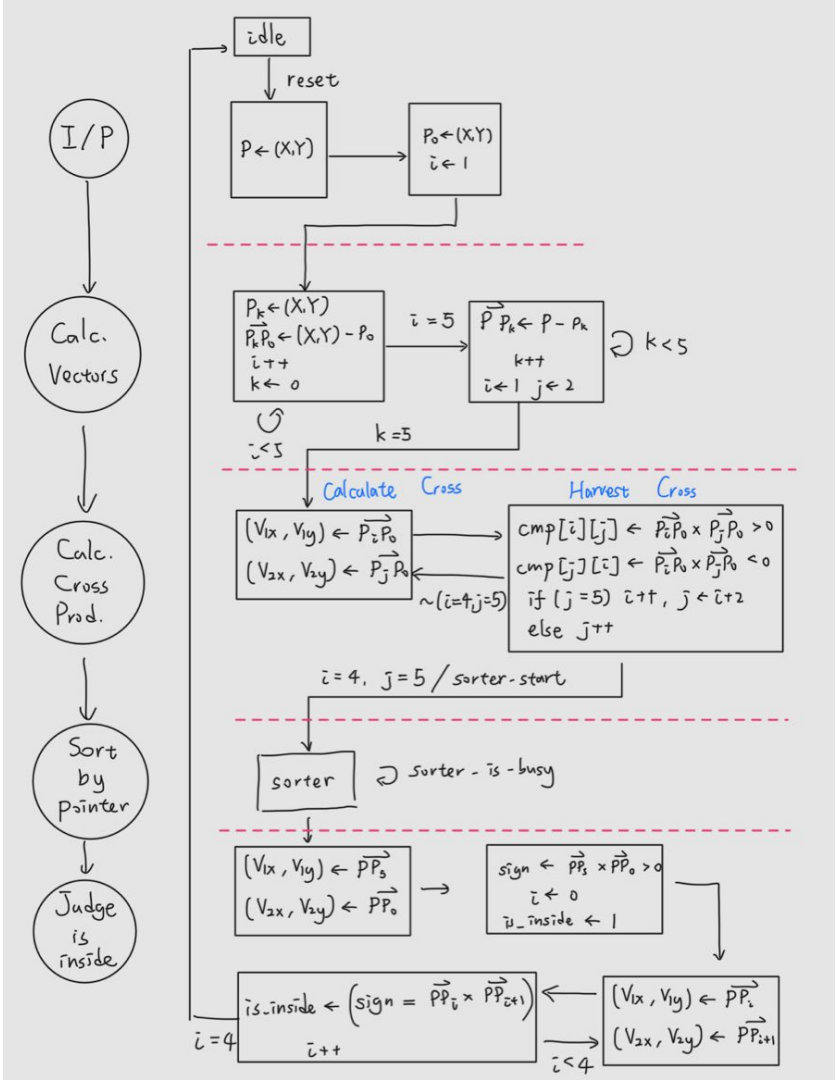




Bitonic Sorting Network



Overall State Machine



Bad-Result

Clock Cycle Time: 30 ns

Total Simulation Time: 8115 (ns)

Total cell area: 42886 (μm^2)

AT: 3.48 × 10*9 (ns-μm^2)

Level: B

area.log

```
Combinational area:                27635.369528
Buf/Inv area:                       1828.099776
Noncombinational area:             15251.138968
Macro/Black Box area:                  0.000000
Net Interconnect area:            461815.676636

Total cell area:                   42886.508495
Total area:                       504702.185131
```

RTL Simulation Success

```
Scenario50(out):    X     Y
        Object:   321, 1000
           AP1:   321,  870
           AP2:   432,  123
           AP3:   970,  280
           AP4:   567, 1003
           AP5:   321,  127
           AP6:  1010,  643
Object50: Golde/Return => 0/0, PASS

----------------------------------------------
--    Simulation finish,  ALL PASS            --
----------------------------------------------
$finish called from file "tb.sv", line 149.
$finish at simulation time             8115000
           V C S   S i m u l a t i o n   R e p o r t
Time: 81150000 ps
CPU Time:      0.840 seconds;       Data structure size:   0.0Mb
Sun Mar 29 04:49:43 2026
CPU time: .851 seconds to compile + .860 seconds to elab + .572 seconds to link + .890 seconds in simulation
[b13088@cad16 2021_univ_cell]$
```

Gate Level Simulation Success

```
Scenario50(out):    X     Y
        Object:   321, 1000
           AP1:   321,  870
           AP2:   432,  123
           AP3:   970,  280
           AP4:   567, 1003
           AP5:   321,  127
           AP6:  1010,  643
Object50: Golde/Return => 0/0, PASS

----------------------------------------------
--    Simulation finish,  ALL PASS            --
----------------------------------------------
$finish called from file "tb.sv", line 149.
$finish at simulation time            81150000
           V C S   S i m u l a t i o n   R e p o r t
Time: 81150000 ps
CPU Time:      2.120 seconds;       Data structure size:   0.9Mb
Sun Mar 29 04:53:05 2026
CPU time: 2.560 seconds to compile + .981 seconds to elab + .760 seconds to link + 2.173 seconds in simulation
[b13088@cad16 2021_univ_cell]$
```

Problems during implementation

- How to transfer the problem into pure Mealy/Moore machine?
- What is a good template for coding an FSM?
- I guess I should not code bitonic sorting network into a FSM, I should directly design a full combinational circuit instead.
- How to best meld a master FSM and a slave FSM?